

Vaiourec
 Valouro Soluções Tubulares do Brasil S.A.
 Bairro Planalto - Av. Orlino Molitès, 65
 ZIP 30640-010 - Belo Horizonte, MG



MILL TEST CERTIFICATE
 (According to DIN EN 10204.2)
 N°.: 0030056067 / 00
 Sheet 1 / 4

Customer:
 NEO Number: 1221007049

Country:

Inspection: BUREAU VERITAS DO BRASIL SOC CLASSIFICADORA E CERTIFICADORA LTDA

DESCRIPTION OF PRODUCTS: SEAMLESS STEEL PIPE, HOT FINISHED, BEVELED ENDS 37,5 DEG., NORMALIZED

DIMENSIONS: 60,30 mm X 11,07 mm GRADE: X52N # 6

STANDARD: API SPEC 5L, 04 2018, 46TH EDITION - PSL 2

IN ACCORDANCE ALSO TO THE STANDARDS: ASTM A 333M - 18 ASTM A 999M - 18 # ASME SA-333M - 17 ASME SA 999M - 18

CUSTOMER SPECIFICATION: -TS-A333 Gr6/API 5L X52-04, 31.01.2021;

MTR: TPT 10221, Rev.1

SURFACE PROTECTION: EXTERNAL: LACQUER PIPE ENDS PROTECTOR; PLASTIC CAP

TOLERANCES: OUTSIDE DIAMETER (PIPE BODY): -0,45 mm / +0,45 mm WALL THICKNESS: -1,38 mm / +1,66 mm

LENGTH: RANDOM: 10000,00 mm - 11800,00 mm

STANDARD MARKING: PAINT STENCILED IN THE PIPE BODY: MANUFACTURER API SPEC 5L-0150.4 API MONOGRAM MONTH/YEAR ASTM/ASME ASA 333M 60.30 X 11.07 X52N/6 PSL2 SMLS BUREAU VER HEAT

NUMBER LENGTH HF S XXS LT -48C20500 KPA/ND SEQ SEQUENCUMBERS VSB LOGO

SHIPPING MARKING: * P.O 1501092021 IMP *

* MADE IN BRAZIL * 606541200

Heat	Pieces	Length(m)
195757	189	2189,61
Total	189	2189,61

THE PRODUCT IS SATISFACTORY IN THE FOLLOWING TESTS / INSPECTIONS: DIMENSIONAL # VISUAL # FLATTENING TEST # ENDS INSPECTION: UNTESTED ENDS CROPPED # HYDROSTATIC TEST: 20500,0 KPA
 5S # RESIDUAL MAGNETISM: MAX 20 GAUSS # ULTRASONIC TEST: API 5L, N5LONG/EI # ULTRASONIC TEST FOR LAMINAR DEFECTS: ISO10893-8-BACK WALL ECO #

PART# 00-1445-0005-00
 MTR# 195757
 PO TM00190770

TAMING MARTINS
 SUBSISTENTES
 WITNESS
 DATA: 15/09/2021



(According to DIN EN 10204-3.2)
Nº.: 0030056067 / 00

Hardness Test Through Wall
Scale: HV10

		Individual	
		Regular: Min	246.0
		Max	
Heat	Control Lot	MTW1	MTW2
195757	030003750251	Q1 189.0	187.0
030003757151		Q1 165.0	164.0
			162.0
			164.0

MW - Middle Wall;
Impact Test

Test Specimen: CHARPY 10X55X10 V NOTCH

Direction: Longitudinal

Temperature: -48°C

	AE1	AE2	AE3	AE4	AE5	AE AVE
Required: Min	(J)	(J)	(J)	(J)	(J)	(J)
Max	4	4	4			7
Heat Control Lot						
195757 030003750251	111	132	196			146
030003757151	138	171	118			142

AE - Absorbed Energy:

Remarks:

Steel fully Killed and produced using fine grain practices

MATERIAL IN ACCORDANCE WITH NACE MR 0175 PAR. 3.2/ISO 15156 - 2015 - ANNEX A.2.1.2/NACE MR0103 PAR.2.1. - 2015

Steel Making:

During the steel making process, Vacuum degassing of the steel was performed.

NORMALIZED

TEMPERATURE 89.5°C

FURNACE TIME 38 MINUTES

COOLING METHOD: AIR





Vaiourec Soluções Tubulares do Brasil S.A.
Barro Preto - Av. Orlino Moura, 65
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MILL TEST CERTIFICATE

(According to DIN EN 10204.1.2)

Nº.: 0030056067 / 00

Sheet / 4

Customer:
NEO Number: 1221007049

Country:

Inspection: BUREAU VÉRITAS DO BRASIL SOC CLASSIFICADORA E CERTIFICADORA LTDA

DESCRIPTION OF PRODUCTS: SEAMLESS STEEL PIPE, HOT FINISHED, BEVELED ENDS 37,5 DEG., NORMALIZED

DIMENSIONS: 60,30 mm X 11,07 mm GRADE: X52N # 6

STANDARD: API SPEC 5L, 46TH EDITION - PSL 2

IN ACCORDANCE ALSO TO THE STANDARDS: ASTM A 333M - 18 ASTM A 999M - 18 # ASME SA-333M - 17 ASME SA 999M - 19

CUSTOMER SPECIFICATION: -TS-A333 Gr.6/API 5L X52-46, 31.01.2021;

MTR: TTT 10221, Rev.1

SURFACE PROTECTION: EXTERNAL: LACQUER PIPE ENDS PROTECTOR: PLASTIC CAP

TOLERANCES: OUTSIDE DIAMETER (PIPE BODY): -0,45 mm / +0,45 mm WALL THICKNESS: -1,38 mm / +1,66 mm

TOLERANCES: PIPE ENDS: OUTSIDE DIAMETER: -0,40 mm / +0,80 mm

LENGTH: RANDOM 10000,00 mm - 11800,00 mm

STANDARD MARKING: PAINT STENCILED IN THE PIPE BODY: MANUFACTURER API SPEC 5L-030.4 API MONOGRAM MONTH/YEAR ASTM/ASME ASA 333M 60.30 X 11.07 X52N/6 PSL2 SMLS BUREAU VER HEAT

NUMBER LENGTH HF 8 X52 LT 48C 20500 KPA-NDE SEQ. 3580008883 VSB LOGO

SHIPPING MARKING: * P.O. 1501092021 IMP *

* MADE IN BRAZIL * 605541/200

Heat	Pieces	Length(m)
195757	189	2189,61
Total	189	2189,61

THE PRODUCT IS SATISFACTORY IN THE FOLLOWING TESTS / INSPECTIONS: DIMENSIONAL # VISUAL # FLATTENING TEST # ENDS INSPECTION : UNTESTED ENDS CROPPED # HYDROSTATIC TEST: 20500,0 KPA

55 # RESIDUAL MAGNETISM: MAX 20 GAUSS # ULTRASONIC TEST : API 5L, N5 LONG EI # ULTRASONIC TEST FOR LAMINAR DEFECTS: ISO10893-B-BACK WALL ECO #

PART# 00-1445-0005-00
MTR# 195757
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MILL TEST CERTIFICATE
(According to DIN EN 10204.1.2)
Nº.: 0030056067 / 00
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Hardness Test Through Wall
Scale: HV10

Heat	Control Lot	Individual		MW1	MW2	MW3	Avg
		Required: Min	Max				
195757	030003750251		245,0	Q1 189,0	187,0	178,0	185,0
	030003757151			Q1 165,0	164,0	162,0	164,0

MW - Middle Wall:

Impact Test

Test Specimen: CHARPY 10X55X10 V NOTCH
Sinker radius: 8 mm

Direction: Longitudinal

Temperature: -80°C

Heat	Control Lot	Required: Min		Max	AE1	AE2	AE3	AE4	AE5	AE Avg
		(J)	(J)		(J)	(J)	(J)	(J)	(J)	(J)
195757	030003750251									
	030003757151									

AE - Absorbed Energy

Remarks:

Steel fully killed and produced using fine grain practices

MATERIAL IN ACCORDANCE WITH NACE MR 0175 PAR. 3.2/ISO 15156 - 2015 - ANNEX A.2.1/ENACE MR0103 PAR.2.1. - 2015

Steel Making:

During the steel making process, Vacuum degassing of the steel was performed.

NORMALIZED
TEMPERATURE 805°C
FURNACE TIME 38 MINUTES
COOLING METHOD : AIR



vaiourec

Vaiourec Soluções Tubulares do Brasil S.A.
Barraão Plant - Av. Otávio Mello, 65
ZIP 30640-010 - Belo Horizonte, MG



MILL TEST CERTIFICATE
(According to DIN EN 10204.3.2)
Nº.: 0030056067 / 00

Sheet / 4

Customer:
NBO Number: 1221007049

Country:

Inspection: BUREAU VERITAS DO BRASIL SOC CLASSIFICADORA E CERTIFICADORA LTDA

DESCRIPTION OF PRODUCTS: SEAMLESS STEEL PIPE, HOT FINISHED, BEVELED ENDS 37,5 DEG., NORMALIZED

DIMENSIONS: 60,30 mm X 11,07 mm GRADE: X52N # 6

STANDARD: API SPEC 5L 04-2018, 46TH EDITION - PSL 2

IN ACCORDANCE ALSO TO THE STANDARDS: ASTM A 333M - 18 ASTM A 999M - 18 # ASME SA-333M - 17 ASME SA 999M - 19
CUSTOMER SPECIFICATION: -TS-A333 Gr.6/API 5L X52-46, 31.01/2021;

MIP- TPT 10221, Rev.1

SURFACE PROTECTION EXTERNAL: LACQUER PIPE ENDS PROTECTORS: PLASTIC CAP

TOLERANCES: OUTSIDE DIAMETER (PIPE BODY): -0,45 mm / +0,45 mm W ALL THICKNESS: -1,38 mm / +1,66 mm

TOLERANCES PIPE ENDS: OUTSIDE DIAMETER: -0,40 mm / +0,80 mm

LENGTH: RANDOM 10000,00 mm - 11300,00 mm

STANDARD MARKING: PAINT STENCILED IN THE PIPE BODY: MANUFACTURER API SPEC 5L 04-2018 API MONOGRAM MONTH/YEAR ASTM/ASME AISA 333M 60.30 X 11.07 X52N/6 PSL 2 SMILS BUREAU VER HEAT
NUMBER LENGTH HR S XXS LT -48C 20500 KPA/ANDE SEQ SEQ/NUMBERS VSB LOGO

SHIPPING MARKING:

- P.O 1501/09/2021 IMP -

MADE IN BRAZIL - 606541/200

Heat	Pieces	Length(m)
195757	189	2189,61
Total	189	2189,61

THE PRODUCT IS SATISFACTORY IN THE FOLLOWING TESTS / INSPECTIONS: DIMENSIONAL # VISUAL # FLATTENING TEST # ENDS INSPECTION: UNTESTED ENDS CROPPED # HYDROSTATIC TEST: 20500,0 KPA
5 S # RESIDUAL MAGNETISM: MAX 20 GAUSS # ULTRASONIC TEST: API 5L N3 LONG ET # ULTRASONIC TEST FOR LAMINAR DEFECTS: ISO10893-8-BACK WALL EGO #

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Hardness Test Through Wall
Scale: HV10

Required: Min		Individual	
Max		246,0	
Heat	Control Lot	MW1	MW2
195757	030003750251	Q1 189,0	187,0
030003757151		Q1 165,0	164,0
MW - Middle Wall:		162,0	164,0

Impact Test

Test Specimen: CHARPY 10X55X10 V NOTCH
Striker radius: 8 mm
Direction: Longitudinal
Temperature: -48°C

Required: Min		Max	
4		7	
Heat	Control Lot	AE1	AE2
195757	030003750251	(J) 111	(J) 132
030003757151		(J) 138	(J) 171
AE - Absorbed Energy:		118	146

Remarks:

Steel fully killed and produced using fine grain practices

MATERIAL IN ACCORDANCE WITH NACE MR 0175 PAR. 3.2/ISO 15156 - 2015 - ANNEX A.2.1.2/NACE MR0103 PAR.2.1. - 2015

Steel Making:

During the steel making process, Vacuum degassing of the steel was performed.

NORMALIZED

TEMPERATURE 895°C

FURNACE TIME 38 MINUTES

COOLING METHOD : AIR



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DESCRIPTION OF PRODUCTS: SEAMLESS STEEL PIPE, HOT FINISHED, BEVELED ENDS 37,5 DEG., NORMALIZED

DIMENSIONS: 60,30 mm X 11,07 mm GRADE: X52N # 6

STANDARD: API SPEC 5L, 64.2018, 46TH EDITION - PSL 2

IN ACCORDANCE ALSO TO THE STANDARDS: ASTM A 333M - 18 ASTM A 999M - 18 # ASME SA-333M - 17 ASME SA 999M - 19

CUSTOMER SPECIFICATION: -TS-A333 Gr.60/API 5L X52-46, 31.01.2021;

MIP: TPT 10221, Rev.1

SURFACE PROTECTION: EXTERNAL: LACQUER PIPE ENDS PROTECTOR; PLASTIC CAP

TOLERANCES: OUTSIDE DIAMETER (PIPE BODY): -0,45 mm / +0,45 mm W/A 1; THICKNESS: -1,38 mm / +1,66 mm

TOLERANCES: PIPE ENDS: OUTSIDE DIAMETER: -0,40 mm / +0,60 mm

LENGTH: RANDOM: 10000,00 mm - 11300,00 mm

STANDARD MARKING: PAINT STENCILED IN THE PIPE BODY: MANUFACTURER API SPEC 5L-0150.4 API MONOGRAM MONTHLY YEAR ASTM/ASME ASA 333M 60,30 X 11,07 X52N/6 PSL 2 SMLS BUREAU VER HEAT

NUMBER LENGTH HF S X52N LT -48C 20500 KPA NDE SEQ 33EQ NUMBERS VSB LOGO

SHIPPING MARKING:

* P.O. 1501092021 IMP *

* MADE IN BRAZIL * 606541200

Heat	Pieces	Length(m)
195757	189	2189,61
Total	189	2189,61

THE PRODUCT IS SATISFACTORY IN THE FOLLOWING TESTS / INSPECTIONS: DIMENSIONAL # VISUAL # FLATTENING TEST # ENDS INSPECTION : UNTESTED ENDS CROPPED # HYDROSTATIC TEST: 20500,0 KPA
5S # RESIDUAL MAGNETISM: MAX 20 GAUSS # ULTRASONIC TEST : API 5L, NS LONG EI # ULTRASONIC TEST FOR LAMINAR DEFECTS: ISO 10893-8-BACK WALL ECO #

PART# 00-1445-0005-00
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MILL TEST CERTIFICATE
(According to DIN EN 10204.3.2)
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Hardness Test Through Wall
Scale: HV10

Required: Min
Max 245,0

Heat	Control Lot	MV1	MV2	MV3	Avg
195757	030003750251	Q1 189,0	187,0	178,0	185,0
	030003757151	Q1 165,0	164,0	162,0	164,0

MV - Middle Wall

Impact Test

Test Specimen: CHARPY 10X55X10 V NOTCH
Sticker radius: 8 mm

Direction: Longitudinal

Temperature: -48°C

AE1	AE2	AE3	AE4	AE5	AE Avg
(J)	(J)	(J)	(J)	(J)	(J)
Required: Min	4	4	4		7
Max					

Heat	Control Lot	111	132	196	146
195757	030003750251				
	030003757151	138	171	118	142

AE - Absorbed Energy

Remarks:

Steel fully killed and produced using fine grain practices

MATERIAL IN ACCORDANCE WITH NACE MR 0175 PAR. 3.2/ISO 15156 - 2015 - ANNEX A.2.1.2/NACE MR0103 PAR.2.1. - 2015

Steel Making:

During the steel making process, Vacuum degassing of the steel was performed.

NORMALIZED

TEMPERATURE 895°C

FURNACE TIME 38 MINUTES

COOLING METHOD: AIR

